

**PHYLLANTHUS EMBLICA (AMLA) SEEDS BIOMASS
DERIVED HARD CARBON ANODE FOR
METAL ION BATTERIES**



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PHYLLANTHUS EMBLICA (AMLA) SEEDS BIOMASS DERIVED HARD CARBON ANODE FOR METAL ION BATTERIES

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by

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Under the supervision of

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**DEPARTMENT OF CHEMICAL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY, AGARTALA**

May-2026

DEDICATION

I dedicate this project to my Project Supervisor Dr. Suresh Mamidi, Assistant Professor, Department of Chemical Engineering, NIT Agartala, whose invaluable guidance, unwavering support and profound knowledge have been instrumental in the successful completion of this project. His insightful feedback, patience, and encouragement have significantly contributed to the quality and depth of this work.

I also extend my gratitude to all the faculty members of the department for their continuous support and contributions to our professional development.

I am also grateful to all those people who have, in any way, guided and supported me throughout the completion of the project.

APPROVAL SHEET

This thesis entitled “*Phyllanthus emblica* (Amla) seeds biomass derived hard carbon anode for Metal Ion batteries” by **Sudipta Paul (22UCH013)**, is approved for the degree of Bachelor of Technology in Chemical Engineering, National Institute of Technology, Agartala, India.

Examiners

Supervisor

Chairman

Date: _____

Place: _____

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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CERTIFICATE

It is certified that the work contained in the thesis titled “*Phyllanthus emblica* (Amla) seeds biomass derived hard carbon anode for Metal Ion batteries” by Sudipta Paul (22UCH013) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Dr. Suresh Mamidi

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May - 2026

PREFACE

This thesis presents a comprehensive study on the development, characterization, and performance analysis of biomass-derived hard carbon as an anode material for metal-ion batteries, with particular emphasis on lithium-ion and sodium-ion batteries. The work focuses on the utilization of *Phyllanthus emblica* (Amla) seeds biomass as a sustainable and cost-effective precursor for the synthesis of hard carbon.

With the rapid growth in energy demand and the increasing need for efficient energy storage technologies, significant attention has been directed toward the development of advanced electrode materials. Although lithium-ion batteries dominate the current market, concerns related to limited lithium resources and cost have encouraged the exploration of alternative batteries such as sodium-ion batteries. In this context, hard carbon materials derived from biomass offer a promising solution due to their environmental friendliness, low cost, and tunable structural properties.

This study aims to systematically investigate how synthesis conditions influence the structural, morphological, and electrochemical properties of hard carbon. Controlled carbonization at elevated temperatures, followed by chemical purification and electrode fabrication, has been employed to develop a material suitable for energy storage applications. The influence of structural features such as interlayer spacing, pore architecture, and defect density on ion storage behavior has been carefully analyzed.

A combination of characterization techniques, including X-ray diffraction (XRD), Raman spectroscopy, and field emission scanning electron microscopy (FESEM), has been used to examine the structural and morphological properties of the synthesized material. Furthermore, electrochemical performance has been evaluated through electrochemical impedance spectroscopy (EIS), cyclic voltammetry (CV), and rate capability analysis to assess its suitability for lithium-ion and sodium-ion batteries.

The findings of this work demonstrate that biomass-derived hard carbon exhibits a disordered turbostratic structure with a porous morphology, making it suitable for metal-ion storage. The study also highlights the importance of optimizing synthesis parameters to achieve improved electrochemical performance. Overall, this work contributes to the growing field of sustainable energy storage materials and provides insights into the potential application of biomass-derived carbon in next-generation battery technologies.

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Regards

Sudipta Paul (22UCH013)

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NOMENCLATURE

Symbol / Abbreviation	Description
A.S.C	Amla Seed Carbon
A.W.	Acid Washed
FESEM	Field Emission Scanning Electron Microscopy
XRD	X-ray Diffraction
EIS	Electrochemical Impedance Spectroscopy
CV	Cyclic Voltammetry
PVDF	Polyvinylidene Fluoride
NMP	N-methyl-2-pyrrolidone
R_s	Solution Resistance
R_{ct}	Charge Transfer Resistance
d_{002}	Interlayer Spacing
L_c	Crystallite Size (stacking direction)
L_a	In-plane Crystallite Size
I_D/I_G	Intensity Ratio of D-band to G-band
Li^+	Lithium Ion
Na^+	Sodium Ion
$mAhg^{-1}$	Specific Capacity
$mA g^{-1}$	Current Density

ABSTRACT

The increasing global demand for efficient and sustainable energy storage systems has driven significant research toward the development of advanced electrode materials for rechargeable batteries. Lithium-ion batteries (LIBs) are widely used due to their high energy density, long cycle life, and excellent electrochemical performance. However, concerns related to resource availability and cost have encouraged the exploration of alternative systems such as sodium-ion batteries (SIBs), which offer advantages in terms of abundance and low cost.

Among various electrode materials, carbon-based materials, particularly hard carbon, have attracted considerable attention due to their low cost, high availability, and tunable structural properties. Hard carbon derived from biomass precursors is especially promising because it provides a sustainable and environmentally friendly route for material synthesis. The disordered structure, enlarged interlayer spacing, and porous nature of hard carbon make it suitable for both lithium-ion and sodium-ion storage.

The electrochemical performance of hard carbon is strongly influenced by its structural characteristics, including interlayer spacing, degree of graphitization, pore structure, and surface chemistry. Controlled carbonization plays a key role in tailoring these properties. In addition, post-treatment processes such as acid washing help remove inorganic impurities, improve surface purity, and enhance electrochemical stability.

In this work, biomass-derived hard carbon was synthesized from Amla seed precursor through controlled carbonization at 1000 °C, followed by purification and electrode fabrication. The structural and morphological properties were analyzed using X-ray diffraction (XRD), Raman spectroscopy, and field emission scanning electron microscopy (FESEM). The electrochemical performance was evaluated using electrochemical impedance spectroscopy (EIS), cyclic voltammetry (CV), and rate capability analysis in both lithium-ion and sodium-ion batteries.

The objective of this study is to investigate the structural properties and electrochemical behavior of biomass-derived hard carbon and to evaluate its suitability as an anode material for both lithium-ion and sodium-ion batteries.

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1. LITERATURE REVIEW

1.1 Introduction – Why Biomass-Derived Hard Carbon for Metal-Ion Batteries?

Biomass-derived hard carbon has emerged as a promising anode material for metal-ion batteries due to its low cost, sustainability, and abundant availability. Unlike graphite, hard carbon possesses a disordered (turbostratic) structure with enlarged interlayer spacing, which facilitates the storage of larger ions such as sodium and potassium. The inherent porous structure and high defect density provide multiple active sites for ion adsorption and improve ion transport within the electrode. Additionally, the physicochemical properties of hard carbon can be tuned by controlling the precursor type and carbonization conditions, enabling optimization of electrochemical performance. These advantages make biomass-derived hard carbon a highly attractive and scalable material for next-generation energy storage systems.

1.2 Fundamental concepts and storage mechanisms

Hard carbon is a non-graphitizable carbon material with a disordered (turbostratic) structure, enlarged interlayer spacing, and a hierarchical porous network consisting of micropores and macropores. These structural features play a key role in ion storage.

The storage of Li^+ and Na^+ ions occurs through multiple mechanisms. Initially, ions are adsorbed on surface defects, edges, and functional groups present in the carbon structure. Then, ions are inserted between the graphene-like layers due to the enlarged interlayer spacing. In addition, micropores act as active sites for ion storage, while macropores facilitate electrolyte penetration and provide pathways for faster ion transport.

The presence of both micropores and macropores improves overall performance by combining efficient ion storage and enhanced ion diffusion. The disordered carbon structure and defect sites further increase the number of active sites for ion storage.

Overall, the storage behavior in hard carbon is governed by surface adsorption, interlayer insertion, and pore-assisted ion transport, which together determine the electrochemical performance.

1.3 Synthesis of Biomass-Derived Hard Carbon: Routes and Examples

Biomass-derived hard carbon is typically synthesized through thermal and chemical processes using natural precursors such as agricultural waste, fruit peels, and plant-based materials. The most common route involves carbonization (pyrolysis) of the biomass in an inert atmosphere at high temperatures, usually between 700 °C and 1400 °C, which converts organic components into a carbon-rich structure.

Different synthesis routes are used to tailor the structure and properties of the material. Direct pyrolysis is the simplest method, where biomass is heated to form hard carbon with a disordered structure. Pre-treatment methods, such as acid washing or hydrothermal treatment, are often

applied to remove impurities and modify the precursor structure. Chemical activation using agents like KOH or KHCO_3 can further enhance porosity and increase surface area. Additionally, controlled heating rate and temperature play a crucial role in determining interlayer spacing, pore structure, and degree of graphitization.

Various biomass sources such as bamboo, rice husk, banana peel, tea waste, and nut shells have been successfully used to produce hard carbon with desirable properties. These materials demonstrate that the choice of precursor and synthesis conditions significantly influences the final structure and electrochemical performance.

1.4 Biomass Derived Hard-Carbon Anode

I) Preparation of walnut shell-derived hard carbon anode materials for high-performance sodium-ion batteries

Physical Properties:

The biomass-derived hard carbon exhibits a disordered (turbostratic) structure with broad XRD peaks corresponding to the (002) and (100) planes. The optimized sample (HC-2), carbonized at 1000 °C, shows an interlayer spacing of 0.3743 nm, with microcrystallite sizes of $L_a = 3.78$ nm and $L_c = 1.08$ nm. It possesses a specific surface area of 53.16 m^2/g and a hierarchical pore structure dominated by micropores along with closed pores. The pore–pore distance is 4.39 nm, indicating a compact porous network. Raman analysis confirms a high defect density with an ID/IG ratio of 2.96, and TEM images reveal layered carbon structures with an interlayer spacing of 0.37 nm.

Electrochemical Properties:

The optimized hard carbon (HC-2) delivers a reversible capacity of 356 mAh/g at 20 mA/g, with a plateau capacity of 234 mAh/g (65%) and an initial Coulombic efficiency of 92.85%. It exhibits strong rate capability, maintaining capacities of 346, 338, 327, 307, and 280 mAh/g at current densities of 50, 100, 200, 500, and 1000 mA/g, respectively. The material shows good cycling stability with 79.34% capacity retention after 600 cycles at 1000 mA/g. Cyclic voltammetry reveals redox peaks at 0.01 V and 0.1 V, indicating reversible sodium storage, while the b-value of 0.37 confirms a diffusion-controlled mechanism.

II) Buckwheat-Derived Hard Carbon

Physical Properties:

The buckwheat-derived hard carbon exhibits a disordered carbon structure with a porous morphology developed through pyrolysis. The material shows high porosity and tunable pore structure, influenced by carbonization temperature and preoxidation treatment. The precursor composition (cellulose, hemicellulose, and lignin) contributes to the formation of a porous carbon framework, while preoxidation enhances pore development and improves structural characteristics. The optimized sample is obtained after preoxidation followed by carbonization at 1400 °C, which promotes better structural development and pore formation.

Electrochemical Properties:

The optimized sample (preoxidized and carbonized at 1400 °C) delivers a reversible capacity of 330.23 mAhg⁻¹, with Coulombic efficiency of 97–98% and stable cycling performance over 100 cycles, indicating good electrochemical stability and efficient sodium-ion storage behavior.

III) Jackfruit Seed–Derived Hard Carbon:

Physical Properties

The hard carbon derived from jackfruit seeds exhibits a disordered carbon structure with partial graphitization, which increases with carbonization temperature. The material shows a porous structure with microspores, and the pore characteristics vary depending on the synthesis conditions. The presence of naturally occurring nitrogen atoms in the biomass contributes to improved electronic conductivity and additional active sites. Structural analysis indicates that the microstructure and pore architecture strongly influence sodium storage behavior. The optimized sample is obtained after carbonization at 1200 °C, which enhances structural ordering and electrochemical performance.

Electrochemical Properties

The optimized hard carbon (carbonized at 1200 °C) delivers a reversible capacity of 221 mAhg⁻¹, with 98.8% capacity retention over 100 cycles, indicating excellent cycling stability. The material also shows good rate performance due to efficient sodium-ion insertion and extraction mechanisms.

IV) Corncob-Derived Hard Carbon:

Physical Properties

The hard carbon derived from waste corncob exhibits a disordered carbon structure with an enlarged interlayer spacing of greater than 0.37 nm, which is higher than that of graphite and facilitates sodium-ion insertion. The material shows a porous structure formed through carbonization and activation processes. The presence of a disordered layered arrangement and porous framework enhances ion accessibility and storage capability. The material is typically obtained after carbonization at high temperatures (around 1000–1300 °C), which promotes the development of the porous carbon framework and suitable interlayer spacing.

Electrochemical Properties

The corncob-derived hard carbon delivers a high reversible capacity of 311 mAhg⁻¹, with an initial Coulombic efficiency of approximately 80%. Under optimized conditions, the material exhibits a maximum specific discharge capacity of 391 mAhg⁻¹, indicating excellent sodium-ion storage performance and good electrochemical stability.

V) Camphor-Derived Hard Carbon:

Physical Properties

The hard carbon derived from camphor wood exhibits a disordered carbon structure with a relatively low defect concentration at optimized heating conditions. The material shows a porous structure formed through controlled carbonization with a low heating rate, which helps in reducing structural defects. The decrease in defect sites and controlled pore structure contributes to improved structural stability and uniform carbon framework formation. The optimized material is obtained through carbonization at around 1000 °C with a controlled low heating rate, which plays a crucial role in minimizing defects and improving structural characteristics.

Electrochemical Properties

The optimized hard carbon delivers an initial charge capacity of 324.6 mAh g⁻¹ with an initial Coulombic efficiency of 82.8%. The material exhibits good cycling stability with 90.0% capacity retention after 200 cycles at 50 mA g⁻¹, indicating stable sodium-ion storage behavior and improved electrochemical performance.

VI) Bamboo-Derived Hard Carbon:

Physical Properties

The hard carbon derived from bamboo exhibits a disordered carbon structure with randomly oriented microcrystals and nanopores. The material shows an interlayer spacing typically in the range of 0.37–0.40 nm, which is favorable for sodium-ion insertion. The structure consists of short-range ordered graphene-like layers stacked in a disordered manner, forming microcrystalline domains with lengths up to 40 Å and consisting of 2–6 layers. The material also contains a large number of nanopores, including both open and closed pores, which contribute to sodium storage. The hard carbon is typically obtained through carbonization at high temperatures (above 1000 °C), which promotes the formation of the disordered carbon framework and development of nanoporous structure.

Electrochemical Properties

The bamboo-derived hard carbon exhibits good sodium storage performance, delivering a reversible capacity of 303.8 mAh g⁻¹ at 30 mA g⁻¹ with an initial Coulombic efficiency of 83.7%, along with improved cycling stability and electrochemical performance after structural optimization.

VII) Tea Waste-Derived Hard Carbon:

Physical Properties

The hard carbon derived from waste tea leaves exhibits a disordered carbon structure with enlarged interlayer spacing and a well-developed porous morphology. The optimized sample (KHCO₃-GT-700) shows a high specific surface area and a micro/mesoporous structure formed through chemical activation. The material also contains abundant defect sites and improved graphitization characteristics, which enhance electrical conductivity and provide additional active sites for ion

storage. The material is obtained through carbonization at **700 °C**, followed by chemical activation, which promotes the development of porosity and defect-rich structure.

Electrochemical Properties

The optimized carbon (KHCO₃-GT-700) delivers a high reversible capacity of 553 mAh g⁻¹ for lithium-ion batteries and maintains a capacity of 211.5 mAh g⁻¹ after 100 cycles at 0.1 A g⁻¹ for sodium-ion batteries, along with excellent cycling stability up to 2000 cycles.

VIII) Banana Peel–Derived Hard Carbon:

Physical Properties

The hard carbon derived from banana peel exhibits a disordered carbon structure with enlarged interlayer spacing and relatively low surface area. The material shows a surface area in the range of 19–217 m² g⁻¹ and an electrode packing density of 0.75 g cm⁻³. The porous structure consists of micro- and mesopores, with pore sizes centered around 3.7 nm at higher carbonization temperatures. XRD analysis confirms the presence of disordered pseudographitic structure, while the material maintains a relatively dense framework compared to conventional activated carbon. The material is obtained through carbonization at temperatures up to 1100–1400 °C, which promotes pseudographitic ordering and pore development.

Electrochemical Properties

The banana peel–derived hard carbon delivers a reversible capacity of 355 mAh g⁻¹ at 50 mA g⁻¹ for sodium-ion batteries. It maintains a capacity of 221 mAh g⁻¹ at 500 mA g⁻¹ with only 7% degradation after 600 cycles, and 336 mAh g⁻¹ at 100 mA g⁻¹ with 11% degradation after 300 cycles. For lithium-ion batteries, it exhibits a high capacity of 1090 mAh g⁻¹ at 50 mA g⁻¹, demonstrating excellent electrochemical performance and stability.

IX) Rice Husk–Derived Hard Carbon:

Physical Properties

The hard carbon derived from rice husk exhibits a disordered carbon structure with randomly oriented turbostratic nanocrystallites. The material shows an enlarged interlayer spacing and suitable oxygen content, which facilitate sodium-ion storage. The structure consists of a porous morphology formed during pyrolysis, and its microstructure varies with carbonization temperature, influencing pore development and structural ordering. The optimized material is obtained after carbonization at 1300 °C, which enhances interlayer spacing and improves electrochemical performance.

Electrochemical Properties

The optimized sample (RHHC-1300) delivers a high reversible capacity of 372 mAh g⁻¹, with an initial Coulombic efficiency of 66%, along with good cycling stability and rate performance. In full sodium-ion batteries, it exhibits an energy density of 185 Wh kg⁻¹, demonstrating strong electrochemical performance.

X) Almond Shell–Derived Hard Carbon:

Physical Properties

The hard carbon derived from almond shells exhibits a disordered carbon structure with enlarged interlayer spacing and a hierarchical porous morphology consisting of micro- and mesopores. The material is synthesized through hydrothermal treatment followed by pyrolysis at 1000 °C and 1200 °C, which promotes the development of a porous carbon framework and improves structural ordering. The presence of defects and oxygen-containing functional groups further enhances ion adsorption and structural characteristics.

Electrochemical Properties

The optimized sample (ASHC1200) delivers an initial reversible capacity of 426 mAhg⁻¹ with an initial Coulombic efficiency of 84.2%. It exhibits a discharge capacity of 379 mAhg⁻¹ at 0.05 A g⁻¹, retains 217 mAhg⁻¹ at 5 A g⁻¹, and maintains 291 mAhg⁻¹ after 1000 cycles at 1 A g⁻¹, indicating excellent rate capability and long-term cycling stability.

1.5 Advantages and Challenges of Biomass-Derived Hard Carbon

Biomass-derived hard carbon offers several advantages as an anode material for metal-ion batteries. It is low cost and readily available due to the abundance of biomass precursors, making it a sustainable and environmentally friendly material source. The disordered carbon structure with enlarged interlayer spacing is particularly suitable for the storage of Li⁺ and Na⁺ ions. In addition, its properties can be tuned by controlling carbonization temperature and treatment methods. The presence of micro- and macropores enhances ion storage and transport, contributing to good electrochemical stability and cycling performance. Furthermore, the synthesis process is relatively simple and scalable, which makes it attractive for large-scale applications.

However, there are several challenges associated with biomass-derived hard carbon. The structure is often irregular and non-uniform due to the natural variability of biomass precursors, leading to limited control over pore size distribution and surface properties. The electrical conductivity is generally lower compared to highly graphitized carbon materials. Additionally, initial irreversible capacity loss occurs due to side reactions and the formation of the solid electrolyte interphase (SEI). Although macropores improve ion transport, they contribute less to charge storage. Moreover, the electrochemical performance strongly depends on synthesis conditions, requiring careful optimization to achieve a balance between porosity, conductivity, and stability.

1.6 Conclusion of the Literature Review

The literature review highlights that biomass-derived hard carbon is a promising and sustainable anode material for both lithium-ion and sodium-ion batteries. Its disordered (turbostratic) structure, enlarged interlayer spacing, and hierarchical porosity make it suitable for efficient ion storage and transport. Various biomass sources such as walnut shell, buckwheat, jackfruit seed, corncob, camphor, bamboo, tea waste, banana peel, rice husk, and almond shell have been successfully utilized to synthesize hard carbon with desirable structural and electrochemical properties.

The reviewed studies demonstrate that synthesis conditions, particularly carbonization temperature and pre-treatment methods, play a crucial role in determining the structural characteristics such as interlayer spacing, pore distribution, and defect density, which directly influence electrochemical performance. Hard carbon materials show good reversible capacity, cycling stability, and rate capability in both Li^+ and Na^+ systems.

However, challenges such as irregular structure, limited control over porosity, and initial irreversible capacity loss still exist. Therefore, further optimization of synthesis methods and structural tuning is required to achieve improved performance.

Overall, biomass-derived hard carbon offers a cost-effective, environmentally friendly, and scalable solution for next-generation energy storage applications, making it a strong candidate for future research and development.

2. SYNTHESIS OF BIOMASS DERIVED HARD CARBON

2.1. Initial Drying of Precursor

(a) Sun Drying

Fresh *Phyllanthus Embilca* seeds biomass was first dried in natural sunlight for 24 hours. This initial drying step removes excess moisture from the seed matrix. It reduces the load for the next thermal drying step. Sun drying also helps prevent microbial residue and biomass damage during vacuum treatment. This ensures a stable precursor for carbonization.



Fig.1- Sun-dried *Phyllanthus emblica* seeds

(b) Vacuum Oven Drying

The sun-dried seeds were placed in a vacuum oven at 80 °C for 24 hours. This step removes deeply trapped water molecules in the lignocellulosic channels and internal pores. It ensures even carbonization. Vacuum drying prevents rapid steam expansion during pyrolysis, which could crack or distort the precursor structure. It also improves consistency in pyrolysis yield.



Fig.2- Vacuum Oven

2.2. Pyrolysis (Carbonization)

Dried seeds were placed into a crucible and then put inside a tubular furnace. Argon gas was used to keep the environment inert and prevent oxidative combustion during heating.

(a) Carbonization at 900 °C

The furnace temperature was increased to 900 °C at a controlled rate of about 3 to 5 °C per minute. During this process, volatile substances such as CO₂, CO, H₂O vapor, hydrocarbons, and tar were released. The organic matrix underwent thermal decomposition, converting macromolecules into char and subsequently into amorphous aromatic carbon. This process results in the formation of highly disordered, turbostratic carbon sheets and also leads to the development of closed pores, which are favorable for Na⁺ and Li⁺ ion storage through pore-filling mechanisms.

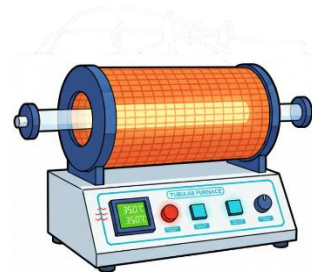


Fig.3- Tubular furnace

(b) Carbonization at 1000 °C

A second batch of the precursor was subjected to the same pre-treatment process, followed by carbonization at 1000 °C under identical conditions. The increase in carbonization temperature leads to improved structural ordering of the carbon matrix due to enhanced rearrangement of carbon layers. As a result, there is a slight reduction in defect density and an increase in the degree of graphitic domains. Additionally, the higher temperature contributes to the development of a more stable and rigid pore-wall structure, which enhances the overall structural integrity of the material.



Fig.4- Carbonized *Phyllanthus emblica* seeds after pyrolysis

2.3 Yield Calculation of Hard Carbon

The yield of hard carbon from *Phyllanthus emblica* seed biomass was measured at two carbonization temperatures: 900 °C and 1000 °C. The carbon yield shows the amount of solid carbon left after pyrolysis compared to the initial mass.

a) Yield at 900 °C

Mass before pyrolysis, $m_{initial} = 11.43 \text{ g}$

Mass after pyrolysis, $m_{final} = 4.25 \text{ g}$

$$\text{Yield (\%)} = \frac{m_{final}}{m_{initial}} \times 100 = \frac{4.25}{11.43} \times 100 = 37.1\%$$

Yield of hard carbon at 900 °C = 37.1%.

b) Yield at 1000 °C

Mass before pyrolysis, $m_{initial} = 15.9 \text{ g}$

Mass after pyrolysis, $m_{final} = 5.8355 \text{ g}$

$$\text{Yield (\%)} = \frac{5.8355}{15.9} \times 100 = 36.6\%$$

Yield of hard carbon at 1000 °C = 36.6 %.

Interpretation:

The yield of hard carbon obtained from A.S. biomass shows a slight decrease with an increase in carbonization temperature. At 900 °C, the yield is 37.1%, whereas at 1000 °C, it decreases to 36.6%.

This reduction in yield at higher temperature can be attributed to the increased decomposition of volatile components such as cellulose, hemicellulose, and lignin during pyrolysis. As the temperature increases, more volatile matter and gases (e.g., CO, CO₂, CH₄) are released, resulting in a lower solid carbon residue.

The relatively small difference in yield indicates that most of the decomposition occurs below 900 °C, and further increase in temperature mainly contributes to structural rearrangement and carbon ordering rather than significant mass loss. Therefore, carbonization at 1000 °C provides improved structural properties with only a marginal reduction in yield.

2.4 Size Reduction (Grinding)

After pyrolysis, the hard carbon was ground mechanically using a clean ceramic mortar and pestle until a uniform, fine powder formed. This grinding step is crucial for preparing electrode-grade carbon. It directly affects coating uniformity, slurry flow, and electrochemical performance.

2.4.1 Purpose and Importance

I) Increase in Effective Surface Area

Grinding breaks down larger carbonized seed fragments into micron-scale powders. This significantly boosts the total active surface area available for ion interaction. As a result, it improves contact between the electrode and electrolyte during cycling and creates better ion-access pathways.

II) Enhanced Slurry Homogeneity

Finer particles mix more uniformly with PVDF and Super P in NMP solvent. A homogeneous slurry reduces sedimentation and maintains consistent viscosity, which leads to stable coating behavior.



Fig.5- Grinding of hard carbon using mortar and pestle

III) Improved Binder Activation and Particle Wetting

Smaller particle size allows for stronger binder adhesion to carbon surfaces. This enhances the mechanical integrity of the electrode film, ensuring that no powder detaches during charge-discharge stress.

IV) Uniform Spreading on Copper Foil

Fine powders create smooth coating layers when applied with a Doctor Blade. They prevent thickness variations and provide even slurry distribution, which is essential for reliable electrochemical results.

V) Minimization of Electrode Cracking and Delamination

Larger particles can cause local stress points during drying and repeated cycling. This may lead to surface cracks, binder failure, and delamination. Micro-sized ground carbon maintains structural compactness, reducing internal voids and improving cycling stability.

VI) Improved Electronic Percolation Network

Smaller particles foster closer contact between the conductive additive (Super P) and carbon surfaces. This forms a continuous conductive pathway, reducing charge-transfer resistance and enhancing high-rate cycling performance.

VII) Better Pore Accessibility

Grinding reveals closed internal pore spaces created during pyrolysis. This increases the accessible micro and meso-porosity and provides more active sites for Li^+/Na^+ storage during adsorption and intercalation processes.

2.5 Acid Treatment with Hydrochloric Acid (HCl)

After reducing the particle size, the powdered hard carbon underwent chemical purification using 1 M hydrochloric acid (HCl). The carbon powder was mixed into the acid solution and kept on a magnetic stirrer at 80 °C and 500 rpm to ensure it stayed suspended and interacted with the fluid. Higher temperatures speed up dissolution, making it easier to remove inorganic residues.

2.5.1 Purpose and Functional Role of Acid Treatment

I) Removal of Residual Mineral Impurities

During biomass growth and pyrolysis, the precursor seeds retain trace mineral impurities such as Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Fe^{3+} , and silicate-based ash. These impurities can disturb the stability at the electrode–electrolyte interface, promote unwanted side reactions, and contribute to irreversible capacity loss. Therefore, their removal is essential to improve the purity and electrochemical performance of the hard carbon material.



Fig.6- Acid Treatment (HCl)

II) Ash Content Reduction

Biomass-derived carbons contain inorganic ash after pyrolysis, primarily originating from natural minerals absorbed by the plant. Acid extraction helps in dissolving these ash components, thereby reducing non-conductive residues and increasing the purity of the carbon material. This process also improves electronic pathways within the electrode matrix, leading to enhanced electrochemical performance.

III) Surface Cleaning and Pore Activation

Acid washing effectively opens buried micropores by dissolving inorganic particles that block pore openings or internal cavities. As a result, the accessible pore volume increases, the carbon surface area improves, and ion transport channels become more defined, facilitating better ion accessibility.

IV) Enhanced SEI Stability

Residual metal ions present in carbon can react with electrolyte salts and solvents, leading to the formation of unstable solid electrolyte interphase (SEI) layers. Their removal reduces irreversible capacity loss during the initial cycle and promotes the formation of a thinner and more uniform SEI layer, resulting in improved Coulombic efficiency in both Li^+ and Na^+ systems.

V) Improved Electrochemical Homogeneity

Purification of the carbon material leads to more uniform electrochemical behavior, which is reflected in smoother voltage profiles, lower impedance response, and reduced fluctuations in both plateau and sloping regions.

VI) Mechanistic Advantage of HCl Purification

Before acid treatment, grain boundaries and pore walls are often partially blocked by metal oxides and mineral residues. After treatment, the removal of these impurities widens pore channels, exposes additional adsorption sites, and enables faster ion transport through the carbon structure, thereby enhancing overall electrochemical performance.

2.6 Acid Washing / Neutralization

After treating with hydrochloric acid, the resulting suspension underwent careful filtration using a funnel and high-quality filter paper. This process separated the solid carbon from the acidic liquid. The hard carbon leftover was washed several times with de-ionized (DI) water to eliminate any remaining chloride ions and dissolved metal salts.

The washing continued systematically until the pH of the filtered liquid stabilized around 7. This indicated complete neutralization and the removal of leftover acid. Achieving a neutral pH is crucial. Any remaining acidity can change the electrode surface chemistry during cycling. This can lead to unwanted side reactions, electrolyte breakdown, and unstable solid electrolyte interphase (SEI) formation.



Fig.7- Acid Washing

2.6.1 Purpose and Importance of Neutralization

I) Complete Removal of Acidic Residues

Neutralization ensures the complete removal of residual hydrochloric acid from the carbon surface. Any remaining acid can promote electrolyte decomposition, corrode cell components, and initiate parasitic reactions during electrochemical cycling. Therefore, thorough washing is essential to maintain chemical stability and ensure reliable electrochemical performance.

II) Elimination of Soluble Chloride Salts

During acid treatment, metal chlorides such as CaCl_2 , MgCl_2 , and KCl are formed. These soluble salts must be completely removed, as their presence can lead to ionic contamination, alter the

electrode potential, and disrupt the formation of a stable solid electrolyte interphase (SEI), ultimately affecting electrochemical performance.

III) Ensuring SEI Stability

Residual acid on the carbon surface can interfere with the formation of a stable and uniform solid electrolyte interphase (SEI). Proper neutralization removes these acidic species, enabling controlled electrolyte decomposition and the formation of a thin, consistent SEI layer. This helps reduce irreversible capacity loss during the initial cycle and improves long-term electrochemical stability.

IV) Prevention of Binder–Carbon Incompatibility

The PVDF binder used in electrode fabrication is sensitive to acidic conditions. Residual acid on the carbon surface can weaken polymer adhesion and lead to non-uniform coating. Proper neutralization ensures good compatibility between the binder and carbon, resulting in uniform electrode films and improved mechanical stability during cycling.

V) Pore Chemistry Restoration

Acid residues trapped within micropores can hinder ion movement and block active sites. Thorough washing removes these residues, fully opening the pore channels and restoring pore accessibility and ionic mobility, thereby improving overall electrochemical performance.

VI) Scientific Rationale

Neutralization ensures that the carbon surface returns to a chemically stable state after acid treatment. Deionized (DI) water is used for washing to prevent the introduction or re-deposition of ionic contaminants during rinsing. Gradual stabilization of the pH to around 7 confirms the complete removal of protonated surface groups and residual chloride ions, indicating effective purification of the material.

Outcome

After thorough washing with deionized water and filtration, the hard carbon powder becomes chemically purified and free from residual impurities. The surface functionality stabilizes, the pore structure remains open and accessible, and the material becomes suitable for final drying and subsequent electrode slurry preparation.

2.7 Drying

After completing the acid washing and achieving a neutral pH, the purified hard carbon powder was subjected to a two-step drying process. Initially, the material was dried at room temperature, allowing surface-bound water to evaporate naturally. This step helps in removing loosely attached moisture without introducing thermal stress to the pore structure immediately after neutralization.

Following air drying, the carbon sample was placed in a vacuum oven and dried under reduced pressure. This process removes moisture trapped within micropores and between turbostratic carbon layers. Vacuum drying ensures complete dehydration, prevents re-adsorption of atmospheric humidity, and minimizes the risk of oxidation that may occur at elevated temperatures in open air.



Fig.8- Dried hard carbon powder after vacuum drying

Outcome

After completing both room-temperature and vacuum drying, the hard carbon powder becomes fully dehydrated, structurally intact, and chemically stable, making it suitable for subsequent slurry preparation and electrode fabrication.

3. ELECTRODE FABRICATION

3.1 Slurry Preparation

To prepare the anode slurry, we used 80 mg of purified hard carbon, 10 mg of Super P, and 10 mg of PVDF. These three components were combined and mixed well in a grinder for even dry blending. This ratio ensures that hard carbon is the main active material storing metal ions. Super P helps with electron transport, and PVDF gives the electrode layer structural strength.

After achieving proper dry mixing, we added N-methyl-2-pyrrolidone (NMP) slowly, drop by drop, to prevent sudden swelling of the binder or uneven wetting. The gradual addition of solvent is important. It lets PVDF dissolve slowly and evenly coat the hard carbon and Super P particles. Mix the slurry continuously until it became a smooth, thick, and highly viscous black paste



Fig.9- Schematic representation of slurry preparation

In this formula, the 80 mg of hard carbon functions as the active ion storage medium, accommodating Li^+ and Na^+ through various methods. The 10 mg of Super P enhances electron movement by creating fine conductive pathways throughout the electrode layer. This reduces resistance and improves charge transport during cycling. The 10 mg of PVDF, fully dissolved in NMP, binds the composite tightly and provides good adhesion when applied to the copper foil, preventing delamination, cracking, or powder detachment during charge and discharge.

The slurry was ready when it had a smooth consistency suitable for doctor-blade coating, showing no sedimentation, binder-rich areas, or thick clumps. At this stage, a stable carbon-binder-conductive network had formed, allowing for a smooth casting in the next step to create a uniform hard-carbon electrode film on copper foil.

3.2 Electrode Coating on Copper Foil

The prepared hard-carbon slurry was placed on a clean copper foil that acted as the current collector. Before coating, the copper substrate was thoroughly cleaned with ethanol or acetone to remove oils, oxide films, and dust particles. This cleaning ensured a strong bond between the slurry and the metal surface. After the foil dried and was free of contaminants, the slurry was poured along the upper edge of the foil and spread using a Doctor Blade. The blade gap was adjusted carefully to control the wet-film thickness. This allowed the slurry to flow evenly across the foil surface without streaking or variation in thickness. This step is vital because the blade height affects the loading mass, electrode porosity, diffusion pathways, and ultimately the electrochemical response of the electrode.



Fig.10
Electrode
Coating

While spreading, the slurry created a smooth, even layer as the Doctor Blade moved at a steady speed. This ensured that hard carbon, Super P, and PVDF were spread evenly across the copper sheet. Proper consistency of the slurry was crucial at this point, as it stopped particle segregation, areas rich in binder, or localized carbon buildup. Once coated, the slurry formed a continuous, defect-free conductive film, firmly held by PVDF and supported by the conductive network from Super P. The resulting electrode sheet had a uniform matte texture, a cohesive binder network, and intact carbon distribution, with no visible cracks, pinholes, or surface problems. This consistency ensured electrical continuity between the carbon layer and the copper current collector, allowing for efficient electron transfer during charge and discharge cycles once assembled into the coin cell.

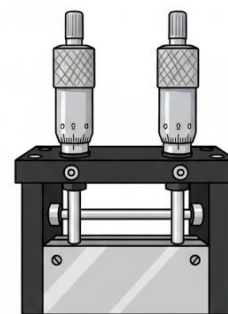


Fig.11 Doctor Blade

3.3. Drying of the Coated Electrode

After applying the hard-carbon slurry evenly to the copper foil, the coated electrode sheet was first left to dry at room temperature. This step allowed most of the NMP solvent to evaporate slowly. This ambient drying phase is important because gradual solvent release prevents internal stress in the electrode layer. Quick drying could cause surface cracks, separation of the binder, or clumping of particles. Once the visible solvent had evaporated and the film became semi-solid, the electrode was moved to a vacuum oven for complete solvent removal. In the oven, reduced pressure and controlled temperature removed any remaining NMP trapped in the pore network or under the binder without degrading the carbon structure or the PVDF binder.

Removing all the solvent is crucial for the mechanical and electrochemical stability of the electrode. Full drying ensures strong, stable adhesion of the hard-carbon layer to the copper current collector. This supports efficient electron transport during cycling and stops delamination during charge and discharge operations. Additionally, having no leftover solvent prevents unwanted chemical reactions with the electrolyte, reduces swelling of the binder, and keeps the designed porosity and microstructure of the electrode film intact. Therefore, the drying process completes the electrode formation, making it structurally stable, chemically clean, and ready for punching and assembly into coin-cell configuration.

3.4. Electrode Cutting and Preparation for Cell Fabrication

After thoroughly drying the coated copper foil and removing all solvents, we prepared the electrode film for the electrochemical cell assembly. We placed the coated sheet on a clean, flat surface and cut circular electrode discs using a precision punching device. Each disc had a uniform diameter to ensure that all assembled cells had the same active material area. This uniformity is

important, as even slight variations in electrode size can affect current density distribution, capacity calculations, and overall electrochemical performance.

During the punching process, we applied steady and controlled force so that the cutting blade passed cleanly through the coated film without tearing, lifting, or deforming the hard-carbon layer. Excessive mechanical stress during cutting can lead to micro-cracks in the coating or partial detachment of the carbon layer from the copper substrate, which could later result in delamination during cycling. We paid special attention to prevent binder fracture at the edges since the circumferential area of the electrode often faces the highest interfacial stress during ion insertion and extraction.

After cutting, we visually inspected each disc against light to ensure smooth and continuous coating along the edges, checking for any flaking, powder separation, or loosely attached carbon fragments. We discarded any electrode that showed coating irregularities or edge damage at this stage to avoid inconsistencies in electrochemical response. The acceptable discs represent fully formed anodes, with a uniform and well-adhered hard-carbon layer bonded to the copper current collector, ready to support stable charge transport during testing.



Fig.12- Punching of coated electrode for coin cell fabrication

To shield the electrodes from atmospheric humidity, oxygen, and fine airborne contaminants, we stored the cut electrodes immediately in an airtight, moisture-free container. Carbon materials, especially after purification and drying, are highly sensitive to ambient adsorption. This can introduce trace water molecules into pores and active sites, negatively impacting initial Coulombic efficiency, altering SEI chemistry when the electrolyte is added, and increasing gas evolution during initial charging. Therefore, keeping the dried discs in a controlled storage environment was vital to maintain their pore structure, surface cleanliness, and electrochemical integrity.

We kept the stored electrodes sealed until it was time to transfer them into the cell assembly workspace. This transfer usually occurs in an inert atmosphere to prevent any additional surface reactions or contamination before electrolyte contact. At this point, the hard-carbon coated discs were fully prepared to function as the active anode components during the fabrication of lithium-, sodium- ion test cells, allowing for direct evaluation of their comparative electrochemical storage performance.

4. RESULT AND DISCUSSION

4.1. X-Ray diffraction & Raman spectra

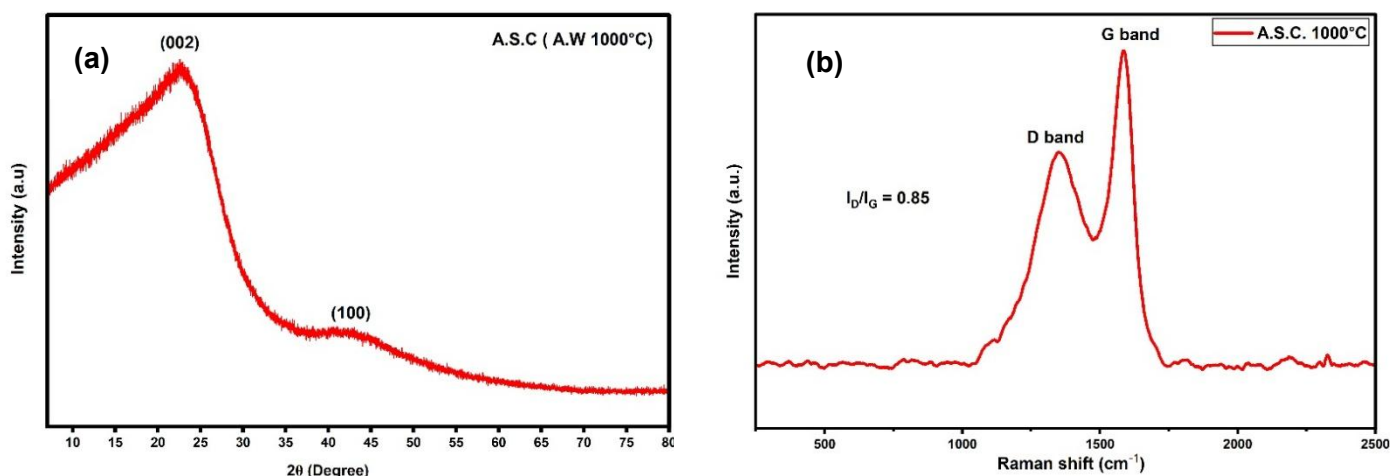


Fig. 13- (a) XRD patterns of A.S.C. (A.W. 1000 °C) (b) Raman spectra of A.S.C. 1000 °C.

The X-ray diffraction (XRD) pattern of the A.S.C. (A.W. 1000 °C) sample carbonized at 1000 °C is shown in Fig. 13 (a). The pattern has a broad peak centered at $2\theta \approx 22.5644^\circ$, which matches the (002) plane of carbon. The broadness of this peak suggests that the carbon structure is disordered and lacks long-range crystalline order.

The interlayer spacing (d_{002}) calculated using Bragg's law is about 0.39 nm. This indicates that the carbon layers are loosely packed. The increased spacing likely results from structural disorder and incomplete graphitization during carbonization. Along with the (002) peak, a weak and broad peak appears at $2\theta \approx 43.04^\circ$, relating to the (100) plane of carbon. This peak shows the in-plane arrangement of carbon atoms. Its low intensity and broadness suggest poor in-plane crystallinity and confirm that only short-range ordering exists within the carbon structure.

The crystallite size (L_c) was found to be approximately 1.42 nm, indicating the presence of very small graphitic domains.

The lack of sharp, well-defined diffraction peaks in the XRD pattern shows that the material is mostly amorphous. Overall, the XRD results show that the synthesized carbon has a disordered turbostratic structure with limited structural ordering.

The Raman spectrum of the synthesized A.S.C sample, carbonized at 1000 °C, is shown in Fig.13 (b). The spectrum exhibits two prominent peaks at 1349.49 cm^{-1} and 1584.42 cm^{-1} , corresponding to the D band and G band, respectively. The D band indicates the presence of structural disorder

and defects, while the G band represents graphitic carbon domains. The presence of both bands confirms that the material consists of both disordered and graphitic regions.

The intensity ratio (I_D/I_G) is found to be approximately 0.85, indicating significant disorder along with partial graphitic ordering. The in-plane crystallite size (L_a) was calculated using the empirical relation $L_a = 4.4 (I_G/I_D)$, and was found to be approximately 5.18 nm, suggesting the presence of small graphitic domains within the disordered carbon matrix.

These results are consistent with the XRD analysis, confirming that the material possesses a disordered turbostratic carbon structure.⁰

4.2. Morphology Analysis (FESEM)

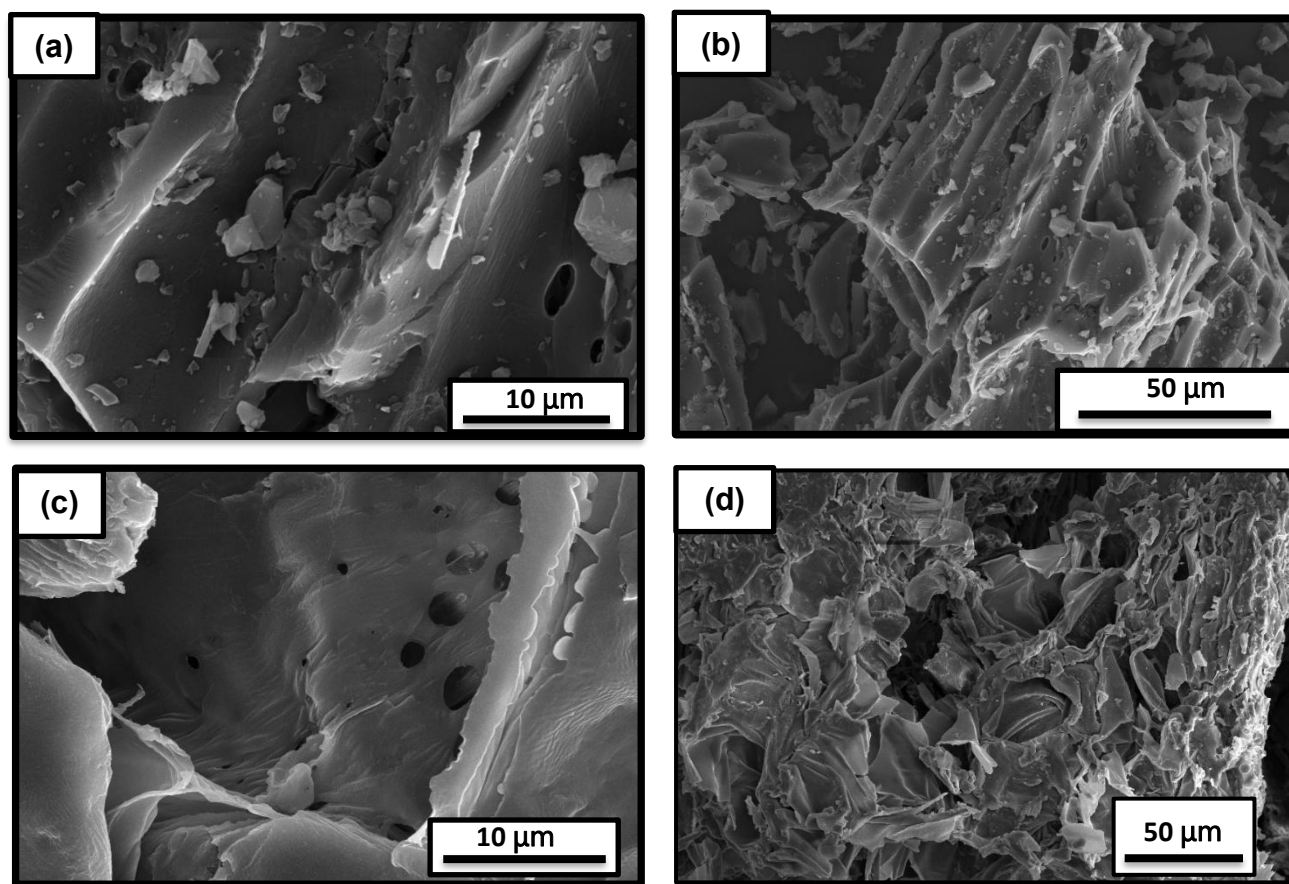


Fig. 14 FESEM images of A.S.C (A.W.) sample: (a) carbonized at 1000 °C (10 μm), (b) carbonized at 1000 °C (50 μm), (c) precursor (10 μm), and (d) precursor (50 μm).

The morphological characteristics of the A.S.C sample carbonized at 1000 °C were characterized using field emission scanning electron microscopy (FESEM). The micrograph reveals a highly porous and irregular surface morphology with a disordered carbon framework. The structure consists of rough, non-uniform, and layered features along with interconnected pore networks, indicating the formation of a heterogeneous carbon structure during the carbonization process.

The pore size, estimated from Fig. 14 (a), was found to be in the range of approximately 215–3719 nm (0.215–3.719 μm), confirming the presence of a well-developed macroporous structure. The wide pore size distribution suggests the formation of hierarchical porosity, which can be attributed to the release of volatile components and gases during carbonization, leading to the creation of void spaces within the material.

The observed disordered and porous structure is consistent with the XRD and Raman analyses, which indicate a turbostratic carbon structure with limited crystallinity. Overall, the FESEM results confirm that the synthesized carbon exhibits a highly porous, disordered, and macroporous structure with hierarchical features.

4.3. Electrochemical Performance

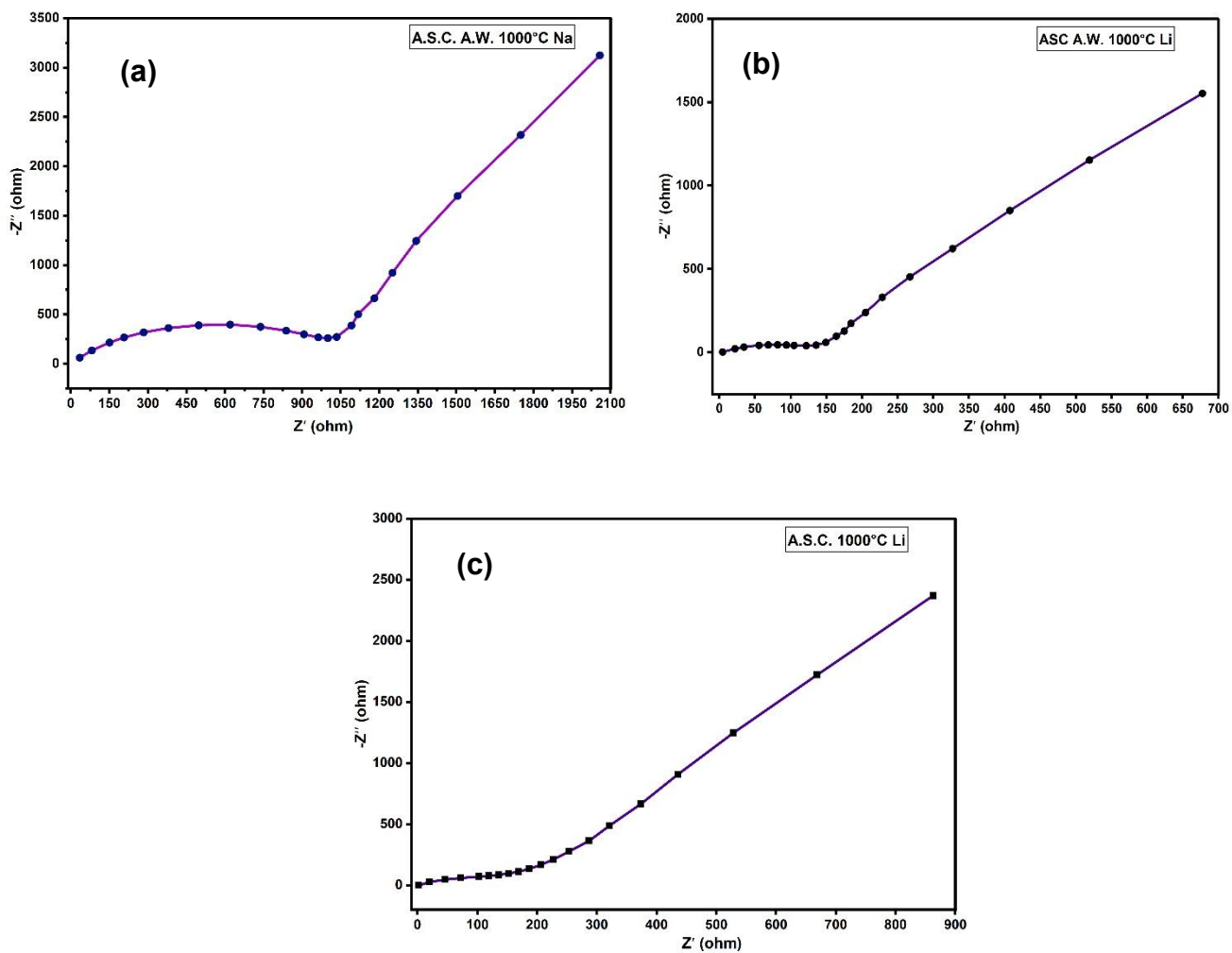


Fig. 15 Electrochemical impedance spectroscopy of (a) ASC A.W. 1000 °C Na-ion (b) ASC A.W. 1000 °C Li-ion (c) A.S.C 1000 °C Li-ion.

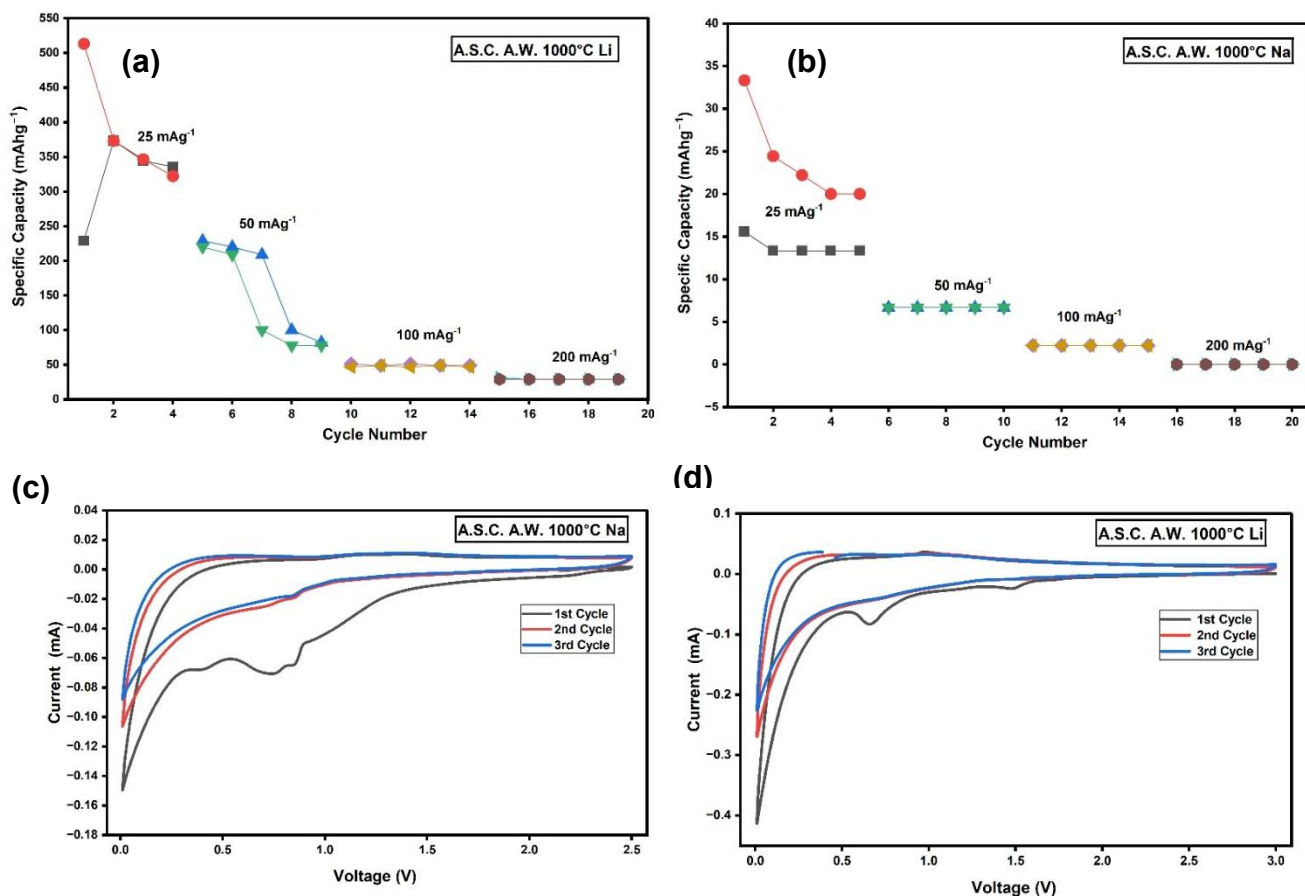


Fig. 16 (a) Rate capability measurement of A.S.C. A.W. 1000 °C Li-ion (b) Rate capability measurement of A.S.C. A.W. 1000 °C Na-ion (c) Cyclic Voltammetry of A.S.C. A.W. 1000 °C Na-ion (d) Cyclic Voltammetry of A.S.C. A.W. 1000 °C Li-ion

The electrochemical properties of the A.S.C sample carbonized at 1000 °C were evaluated using electrochemical impedance spectroscopy (EIS), and the corresponding Nyquist plots are shown in Fig.15(a–c). All plots exhibit a depressed semicircle in the high-frequency region followed by an inclined straight line in the low-frequency region.

For the A.S.C A.W. 1000 °C Na system Fig. 15 (a), the intercept at the real axis corresponds to the solution resistance (R_s), which was found to be 34.49395 Ω , while the charge transfer resistance (R_{ct}) was calculated to be 964.71149 Ω .

For the A.S.C A.W. 1000 °C Li system Fig. 15 (b), the R_s value was determined to be 2.53572 Ω , and the R_{ct} value was calculated to be 119.25755 Ω .

For the A.S.C 1000 °C Li system Fig. 15 (c), a very small semicircle was observed in the high-frequency region, with R_s and R_{ct} values of 1.66681 Ω and 150.56134 Ω , respectively.

The rate capability of the A.S.C A.W. 1000 °C (Li-ion) sample was evaluated at different current densities, as shown in Fig. 16 (a). The electrode delivered specific capacities of 513.33, 228.88, 51.11, and 31.11 mAhg⁻¹ at current densities of 25, 50, 100, and 200 mA g⁻¹, respectively. The rate capability of the A.S.C A.W. 1000 °C (Na-ion) sample is shown in Fig. 16 (b). The electrode delivered specific capacities of 33.33, 6.66, 2.22, and 0 mAhg⁻¹ at current densities of 25, 50, 100, and 200 mA g⁻¹, respectively.

The cyclic voltammetry (CV) curves of A.S.C A.W. 1000 °C for Na-ion and Li-ion systems are presented in Fig. 16(c) and Fig. 16(d), respectively.

In Fig. 16(c), corresponding to the Na-ion system, a broad reduction peak is observed in the first cycle in the voltage range of approximately 0.5–0.9 V, which is attributed to the formation of the solid electrolyte interphase (SEI) layer due to electrolyte decomposition. The peak intensity decreases in the subsequent cycles, indicating partial stabilization of the electrode–electrolyte interface. However, the noticeable difference between the first, second, and third cycles suggests relatively lower reversibility and higher polarization. This behavior is mainly due to the larger ionic radius of Na⁺ ions, which leads to slower diffusion kinetics and limited accessibility to active sites within the hard carbon structure.

In Fig. 16(d), corresponding to the Li-ion system, a similar reduction peak appears in the first cycle around 0.5–0.8 V, indicating SEI formation. In contrast to the Na-ion system, the second and third cycles show a high degree of overlap, demonstrating excellent reversibility and stable electrochemical behavior. The reduced polarization and improved retraceability of the CV curves indicate faster Li⁺ ion transport and better charge transfer kinetics.

For both systems, the sloping region in the voltage range of 0.1–1.5 V corresponds to adsorption and intercalation of ions within the disordered carbon layers, while the low-voltage region (<0.1 V) is associated with pore-filling mechanisms. This combined storage mechanism is characteristic of hard carbon materials with turbostratic structure and hierarchical porosity.

Overall, the A.S.C A.W. 1000 °C sample exhibits superior electrochemical performance in the Li-ion system compared to the Na-ion system, owing to lower polarization, better reversibility, and more efficient ion transport kinetics.

5. CONCLUSION

In this work, biomass-derived hard carbon was successfully synthesized from *Phyllanthus emblica* (Amla) seeds precursor through controlled carbonization at 1000 °C, followed by acid treatment and electrode fabrication. The prepared material was systematically characterized to evaluate its structural, morphological, and electrochemical properties.

X-ray diffraction analysis showed a broad (002) peak at $2\theta = 22.5644^\circ$, confirming the formation of a disordered turbostratic carbon structure. The interlayer spacing (d_{002}) was calculated to be 0.39 nm, and the crystallite size (L_c) was found to be 1.42 nm, indicating small graphitic domains and limited crystallinity. Raman spectroscopy further supported this observation, exhibiting D and G bands at 1349.49 cm^{-1} and 1584.42 cm^{-1} , respectively. The intensity ratio (I_D/I_G) was 0.85, and the in-plane crystallite size (L_a) was calculated to be 5.18 nm, confirming the presence of disordered carbon with partial graphitic ordering.

FESEM analysis revealed a highly porous and irregular surface morphology with pore sizes ranging from 215 to 3719 nm (0.215 to 3.719 μm), confirming the presence of a macroporous and heterogeneous structure.

Electrochemical impedance spectroscopy (EIS) analysis showed that the A.S.C A.W. 1000 °C Na-ion system exhibited an R_s value of 34.49395 Ω and an R_{ct} value of 964.71149 Ω . In contrast, the A.S.C A.W. 1000 °C Li-ion system showed improved electrochemical behavior with R_s and R_{ct} values of 2.53572 Ω and 119.25755 Ω , respectively. The A.S.C 1000 °C Li-ion system exhibited R_s and R_{ct} values of 1.66681 Ω and 150.56134 Ω , respectively.

The rate capability analysis demonstrated that the A.S.C A.W. 1000 °C Li-ion system delivered specific capacities of 513.33, 228.88, 51.11, and 31.11 mAhg^{-1} at current densities of 25, 50, 100, and 200 mA g^{-1} , respectively. In comparison, the Na-ion system delivered significantly lower capacities of 33.33, 6.66, 2.22, and 0 mAhg^{-1} at the same current densities, indicating slower ion kinetics and limited utilization of active sites.

Overall, the synthesized hard carbon exhibits a disordered turbostratic structure with a porous morphology and demonstrates better electrochemical performance in the Li-ion system compared to the Na-ion system.

6. FUTURE SCOPE

The performance of biomass-derived hard carbon can be further improved by optimizing synthesis parameters such as carbonization temperature, heating rate, and precursor treatment conditions. Precise control over these parameters may enhance structural ordering and improve electrochemical properties.

Further enhancement in electrochemical performance can be achieved through heteroatom doping (such as nitrogen, oxygen, or sulfur), which can improve electrical conductivity and introduce additional active sites for ion storage. Chemical or physical activation methods can also be explored to tailor pore structure and enhance ion transport pathways.

Advanced characterization techniques such as BET surface area analysis, TEM, and XPS can be employed to gain deeper insights into pore distribution, surface chemistry, and structural evolution of the material. In addition, optimization of electrode composition, binder system, and electrolyte selection can further improve charge transfer behavior and cycling stability.

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